

Swastik S. Karabashettar



B.Tech student and developer. Applying for the **Tech Squad**, GDG UVCE Core Team 2026-27
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Why the Tech Squad

I learn by building things, and I already end up explaining programming basics, AI/ML concepts, and Git and GitHub to friends and juniors. Running workshops on stuff I've actually used (Python, Next.js, LLM APIs) feels like the natural next step, and I'd like to build bigger things with a team.

Education

University Visvesvaraya College of Engineering (UVCE), Bengaluru
B.Tech., Artificial Intelligence and Data Science

Expected 2029

Projects

CivicPath / NammaPath (government services assistant)

Next.js, TypeScript, Supabase

Helps people in Karnataka figure out which government services they qualify for, what documents they need and how to apply. Works in English, Kannada and Hindi. Covers 15+ state and central services.

Live: government-work-assistant-ai-two.vercel.app

- Eligibility is decided by plain rule logic (condition trees) over a verified database, so fees, documents and links never come from the LLM. The AI only explains the result.
- Tries Groq first and falls back to Gemini, then Claude, if a provider fails. AI answers are checked with Zod before users see them, and it can generate a PDF checklist (pdf-lib).
- Supabase/PostgreSQL with row-level security. 37 Jest tests, 19 commits.

AutoClip (AI short-form video editor)

Python, OpenCV, FFmpeg, Faster-Whisper

Turns long videos into 9:16 shorts automatically. Built on an open-source AutoClip base for the timeline/model parts, plus OpenCV, FFmpeg and Faster-Whisper.

- Finds the speaker's face with OpenCV and moves the vertical crop to follow it, smoothed so it doesn't jitter. Zooms in on high-energy moments.
- Word-by-word karaoke captions from Faster-Whisper, with background music that ducks under speech.
- Every render passes two checks: technical (1080x1920, 30 FPS, loudness, black or frozen frames) and creative (subject visible, pacing, speech gaps). 37 Pytest tests.

Smart Hydroponic Plant Health Monitor (team project of 7)

Scikit-learn, FastAPI, React

Rates a hydroponic plant as Healthy, Warning or Critical with a 0 to 100 score, from sensor readings. I built the software side.

Live: hypotonic-farming-system.vercel.app | Code: github.com/123swastik123/Hypotonic-farming-system

- Made a synthetic dataset of 10,500 readings across lettuce, tomato, spinach, basil and strawberry. The model mostly looks at how far each reading is from that crop's healthy range.
- One classifier for the status and one regressor for the score, served through a FastAPI backend.
- React dashboard with simulated telemetry, charts, suggestions and alerts in English, Hindi and Kannada.

Team project

OmniComm (CodeFury 9.0, team Byte Hogs)

team of 4

- Four-person team building a sign-language communication system. We integrated an existing sign-language recognition model into the app, and I did most of the development and integration work.

Skills

Languages	Python, TypeScript, JavaScript, SQL
AI / ML	Machine learning, computer vision, LLM integration, Scikit-learn, OpenCV, Faster-Whisper
Web / backend	React, Next.js, FastAPI, Tailwind CSS
Data	PostgreSQL, Supabase, SQLite, SQLAlchemy, Pandas, NumPy
Tools	Git, GitHub, FFmpeg, Jest, Pytest; Gemini, Claude and Groq APIs

Also: problem solving, teamwork, communication, attention to detail.